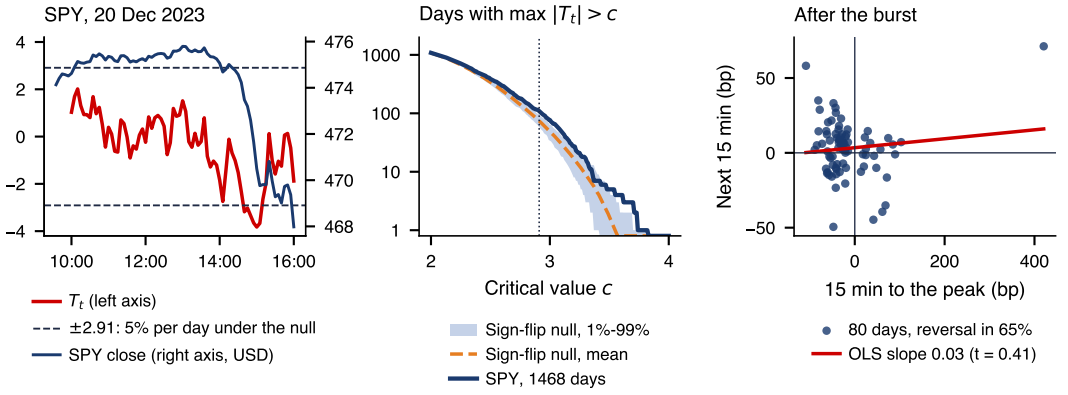
SPY, 20 Dec 2023
4
2
0
−2
−4
476
474
472
470
468
10:00
12:00
14:00
16:00
T_t (left axis)
±2.91: 5% per day under the null
SPY close (right axis, USD)
Days with max $|T_t| > c$
1000
100
10
1
2
3
4
Critical value c
Sign-flip null, 1%-99%
Sign-flip null, mean
SPY, 1468 days
After the burst
Next 15 min (bp)
50
0
−50
0
200
400
15 min to the peak (bp)
80 days, reversal in 65%
OLS slope 0.03 (t = 0.41)